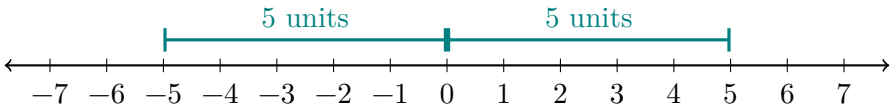
Class 9 Mathematics — Exercise 5.3

Basic Concept

(1) Absolute Value

Absolute value means **"how far a number is from zero"**.



Distance between any two points is always a **positive number or zero**. So the absolute value of a number is always a positive number or zero.

Example: Distance from 0 to 5, or from 0 to −5, is 5 units on the number line.

$$|5| = 5 \qquad |-5| = 5$$

(2) Absolute Value Bars

The two vertical bars | | are called absolute value bars, e.g. $|5|$

(3) Absolute Value Equation

An equation that contains the variable inside the absolute value bars, e.g. $|x+2| = 6$

(4) Key Point

If an absolute value is equal to a **negative number**, then it has **no solution**, because absolute values are always positive.

$$|x+2| = -6 \quad \Rightarrow \quad \text{No Solution}$$

Method used in every question: Remove the bars and write $\pm$ on the other side $\rightarrow$ make two equations ($+$ and $-$) $\rightarrow$ solve both $\rightarrow$ S.S

Question 1

$$|x| = \frac{5}{3}$$

Solution:

$$x = \pm\frac{5}{3}$$

$$\boxed{x = \frac{5}{3}} \quad \Big| \quad \boxed{x = -\frac{5}{3}}$$

$$\boxed{S.S = \left\{\frac{5}{3},\ -\frac{5}{3}\right\}}$$

Question 2

$$|x+2| = 6$$

Solution:

$$x+2 = \pm 6$$

$x+2=6$	$x+2=-6$
$x=6-2$	$x=-6-2$
$\boxed{x=4}$	$\boxed{x=-8}$

$$\boxed{S.S = \{4,\ -8\}}$$

Question 3

$$|5y-1| = 9$$

Solution:

$$5y-1 = \pm 9$$

$5y-1=9$	$5y-1=-9$
$5y=9+1$	$5y=-9+1$
$5y=10$	$5y=-8$
$y=\frac{10}{5}$	$\boxed{y=-\frac{8}{5}}$
$\boxed{y=2}$	

$$\boxed{S.S = \left\{2,\ -\frac{8}{5}\right\}}$$

Question 4

$$|x+1| = 2$$

Solution:

$$x+1=\pm 2$$

$x+1=2$	$x+1=-2$
$x=2-1$	$x=-2-1$
$\boxed{x=1}$	$\boxed{x=-3}$

$$\boxed{S.S=\{1,\ -3\}}$$

Question 5

$$|6-3y|=0$$

Solution:

$$6-3y=0$$
$$-3y=-6$$
$$y=\frac{6}{3}$$
$$\boxed{y=2}$$

$$\boxed{S.S=\{2\}}$$

Question 6

$$3|z-2|-4=-2$$

Solution:

$$3|z-2|=-2+4$$
$$3|z-2|=2$$
$$|z-2|=\frac{2}{3}$$
$$z-2=\pm\frac{2}{3}$$

$z-2=\frac{2}{3}$	$z-2=-\frac{2}{3}$
$z=\frac{2}{3}+\frac{2}{1}$	$z=-\frac{2}{3}+\frac{2}{1}$
$=\frac{2+6}{3}$	$=\frac{-2+6}{3}$

$z-2=\frac{2}{3}$	$z-2=-\frac{2}{3}$
$\boxed{z=\frac{8}{3}}$	$\boxed{z=\frac{4}{3}}$

$$\boxed{S.S=\left\{\frac{8}{3},\ \frac{4}{3}\right\}}$$

Question 7

$$|2x-1|=5$$

Solution:

$$2x-1=\pm 5$$

$2x-1=+5$	$2x-1=-5$
$2x=5+1$	$2x=-5+1$
$2x=6$	$2x=-4$
$x=\frac{6}{2}$	$x=\frac{-4}{2}$
$\boxed{x=3}$	$\boxed{x=-2}$

$$\boxed{S.S=\{3,\ -2\}}$$

Question 8

$$|3x+2|=7$$

Solution:

$$3x+2=\pm 7$$

$3x+2=7$	$3x+2=-7$
$3x=7-2$	$3x=-7-2$
$3x=5$	$3x=-9$
$\boxed{x=\frac{5}{3}}$	$x=\frac{-9}{3}$

$3x + 2 = 7$	$3x + 2 = -7$
	$\boxed{x = -3}$

$$\boxed{S.S = \left\{\frac{5}{3},\ -3\right\}}$$

Question 9

$$\frac{|4x|}{3} = 12$$

Solution:

$$|4x| = 12 \times 3$$
$$|4x| = 36$$
$$4x = \pm 36$$

$4x = 36$	$4x = -36$
$x = \frac{36}{4}$	$x = \frac{-36}{4}$
$\boxed{x = 9}$	$\boxed{x = -9}$

$$\boxed{S.S = \{9,\ -9\}}$$

Question 10

$$|5x| + 10 = 5$$

Solution:

$$|5x| = 5 - 10$$
$$|5x| = -5$$

Absolute value can't be negative, it is always positive. So this equation has no solution.

$$\boxed{S.S = \{\ \} \text{ or } \phi}$$

Question 11

$$\frac{|1-2y|}{4} = 3$$

Solution:

$$|1-2y| = 3 \times 4$$
$$|1-2y| = 12$$
$$1-2y = \pm 12$$

$1-2y = 12$	$1-2y = -12$
$-2y = 12-1$	$-2y = -12-1$
$-2y = 11$	$-2y = -13$
$\boxed{y = -\frac{11}{2}}$	$\boxed{y = \frac{13}{2}}$

$$\boxed{S.S = \left\{-\frac{11}{2}, \frac{13}{2}\right\}}$$

Question 12

$$\frac{|x+1|}{2} = \frac{|2x-1|}{3}$$

Solution:

$$\frac{|x+1|}{|2x-1|} = \frac{2}{3}$$
$$\left|\frac{x+1}{2x-1}\right| = \frac{2}{3}$$
$$\frac{x+1}{2x-1} = \pm\frac{2}{3}$$

$\frac{x+1}{2x-1} = \frac{2}{3}$	$\frac{x+1}{2x-1} = -\frac{2}{3}$
$3(x+1) = 2(2x-1)$	$3(x+1) = -2(2x-1)$
$3x+3 = 4x-2$	$3x+3 = -4x+2$
$3x-4x = -2-3$	$3x+4x = 2-3$
$-x = -5$	$7x = -1$

$\frac{x+1}{2x-1} = \frac{2}{3}$	$\frac{x+1}{2x-1} = -\frac{2}{3}$
$\boxed{x = 5}$	$\boxed{x = -\frac{1}{7}}$

$$\boxed{S.S = \left\{5,\ -\frac{1}{7}\right\}}$$

Question 13

$$|5x - 3| = |x + 7|$$

Solution:

$$5x - 3 = \pm(x + 7)$$

$5x - 3 = x + 7$	$5x - 3 = -(x + 7)$
$5x - x = 7 + 3$	$5x - 3 = -x - 7$
$4x = 10$	$5x + x = -7 + 3$
$x = \frac{10}{4}$	$6x = -4$
$\boxed{x = \frac{5}{2}}$	$x = -\frac{4}{6}$
	$\boxed{x = -\frac{2}{3}}$

$$\boxed{S.S = \left\{\frac{5}{2},\ -\frac{2}{3}\right\}}$$

Question 14

$$|z + 3| - 3 = 5 - |z + 3|$$

Solution:

$$|z + 3| + |z + 3| = 5 + 3$$
$$2|z + 3| = 8$$
$$|z + 3| = \frac{8}{2}$$
$$|z + 3| = 4$$
$$z + 3 = \pm 4$$

$z+3=4$	$z+3=-4$
$z=4-3$	$z=-4-3$
$\boxed{z=1}$	$\boxed{z=-7}$

$$\boxed{S.S=\{1,\ -7\}}$$